

WASP-43b

R-0620

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180218 · created 2026-07-25T12:07:32+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458167.7784 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.012
Transit depth (Rp/Rs)^2	3.14 +/- 0.41 [%]
Orbital Inclination (inc)	81.12 +/- 0.78
Airmass coefficient 1 (a1)	0.9708 +/- 0.008
Airmass coefficient 2 (a2)	0.0196 +/- 0.0075
Scatter in the residuals of the lightcurve fit is	0.82 %
Best Comparison Star	#1 - [323, 75]
Optimal Aperture	3.53
Optimal Annulus	12.14
Transit Duration (day)	0.0488 +/- 0.0032

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.177 ± 0.012 vs 0.1594 (deviation 1.47σ)
Duration fit vs published	0.0488 d vs 0.0512 d (deviation 0.77σ)
Mid-time O–C	-0.3 min
Depth SNR	14.7
Residual scatter	0.82 %

Light curve

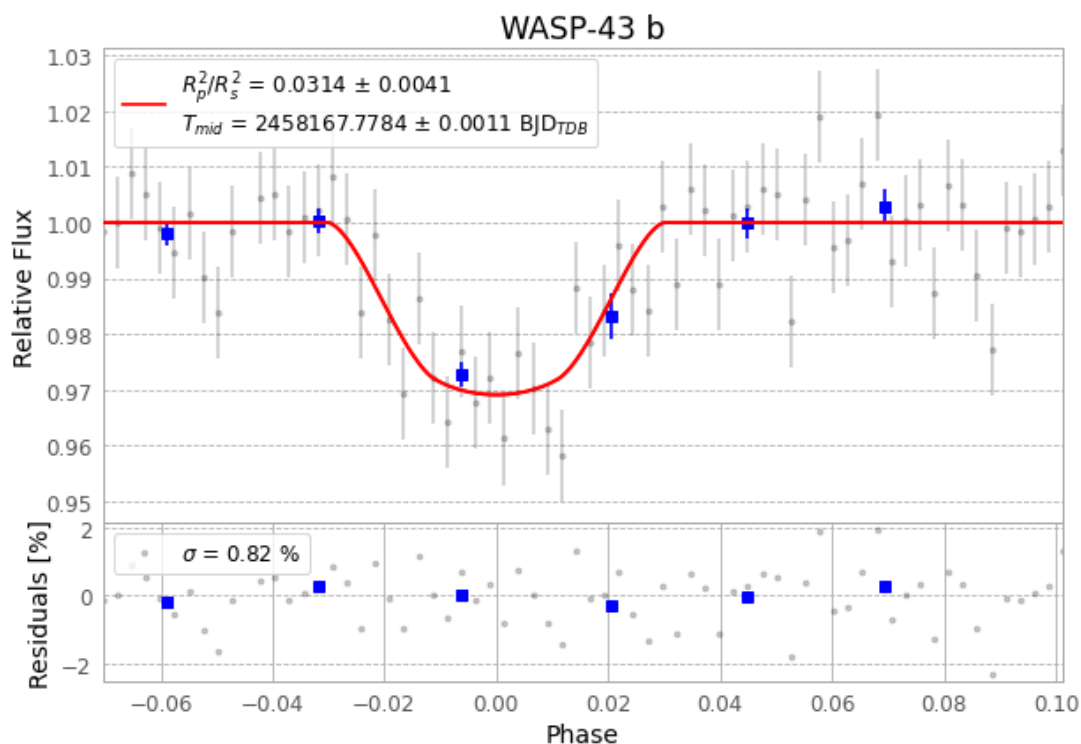


Figure 1 — FinalLightCurve_WASP-43 b_2018-02-17.png (SHA-256 08239c511c88e6c4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1dc37e5b75cd84bb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

68 FITS frames (45 MB), WASP-43180218050912.FITS → WASP-43180218083012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180218.log (f9bc4c978293...), FinalLightCurve_WASP-43 b_2018-02-17.png (08239c511c88...), FinalLightCurve_WASP-43 b_2018-02-17.csv (741f0dc4f3ff...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 12:07Z.